CHAPTER 6 EVIDENCE PACKAGE: CONCLUSIONS, RECOMMENDATIONS AND FUTURE WORK

> **What this file is.** The Chapter 6 entry point of the nine-file handover set: what each conclusion may rest on, the recommendations the evidence supports, and the future work that is genuinely future rather than unfinished. Repository `student-beneficiary-system`, branch `fix/sbms-completion-work`, commit `d6b5032`.
>
> **Full detail:** `CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORKS.md` (§6.1–§6.5), `OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md` §1 and §5, `SBMS_COMPREHENSIVE_CHAPTERS_4_5_6_INFORMATION_REPORT.md` §20–§24.

---

1. The objectives, and where their evidence sits

The four specific objectives are reproduced as working statements below. **The exact Chapter 1 sentences are not in this repository** — Chapters 1–3 exist only outside it — so the author must paste the originals from the report and confirm they match these statements before submission. Chapter 6 must answer the originals, not this file.

#	Working statement	Implemented functionality	Evidence	Extent	Limitation
(i)	Investigate the challenges of manual beneficiary registration and tracking	Requirement and workflow analysis, problem framing	`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md` §4.1–§4.2; proposal §1	Achieved as analysis	No field study or interview record exists
(ii)	Design the architecture, database schema and ML pipeline, incorporating a Random Forest classifier	Three-tier architecture, 13 tables, 14 migrations, six-feature XGBoost pipeline	§4.4, §4.6, §4.7; `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md`	Achieved with a stated deviation	Classifier is XGBoost, not Random Forest (G-21)
(iii)	Implement registration, unique identifiers, QR identification and role-based access control	Registration with validation, generated identifiers, existence-only QR verification, three-role RBAC	`CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION.md` §5.3.2–§5.3.8	Achieved	Physical QR scan never exercised (G-11)
(iv)	Validate, test and evaluate the system's effectiveness	151 backend tests, 46 native tests, 43 browser checks, 40 Python tests, ML evaluation, restore rehearsal	§5.4; `SBMS_TEST_EVIDENCE_CATALOGUE.md`	Partly achieved	No real data, no user acceptance, no load test

2. What the conclusions may and may not say

Conclusion	Supportable wording	Basis
The system is built and behaves as specified	"Route-level and workflow-level behaviour matches the specification across 151 backend tests, 43 browser checks and 40 Python tests, each with its date in §5.4."	Test artefacts with dates
Duplicate detection is functional	"The pipeline scores each candidate pair against the whole registry and surfaces the result to an administrator, who decides."	Code and tests
Duplicate detection is accurate	**Not supportable.** No adjudicated labels; synthetic data only	`SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md` §4
The ML component adds value	"It reduces the reviewer's candidate set to a ranked top-1 per record and stores provenance per flag." Value against a manual baseline is unmeasured	Design evidence only
The system improves on manual processes	**Only as a design argument.** No before-and-after measurement and no users	Gap G-06
The payment chain is controlled	"Payment state is gated in the database as well as the API, and every state change is audited."	Schema constraints, audit tests
The system is production-ready	**Not supportable.** Demonstration instance; no monitoring, backup schedule or penetration test	§5.6.6

3. Recommendations the evidence supports

#	Problem or opportunity	Evidence	Action	Benefit	Implementation consideration
R-1	Labels are synthetic, so accuracy is unknown	ML doc §4, §10	Build an adjudicated ground-truth set from the decisions already stored in `duplicate_review_events`	Real precision, recall and a defensible cut-off	Needs data-sharing permission and a labelling protocol
R-2	Whole-registry comparison is quadratic	Screening service; 22.67 ms at 50 pairs	Add blocking or candidate generation before scoring	Keeps review volume viable as records grow	Blocks on the ML service, not the database
R-3	The score is uncalibrated and unexplained	ML doc §11	Calibrate probabilities and expose the top contributing features per flag	A reviewer can judge a flag instead of trusting it	Feature names must stay stable across retraining
R-4	Flag volume may exceed review capacity	Screening design	Set an explicit flag budget per cycle and measure reviewer throughput	Prevents an unusable queue	Needs an owner decision on the cost of a missed duplicate
R-5	Operation and assurance are thin	`SBMS_DEPLOYMENT_AND_OPERATIONS_GUIDE.md` §9	Add monitoring, a backup schedule and a scheduled restore drill	Reduces operational risk	Costs money; Azure monitoring already available
R-6	Identity assurance rests on a printed token	QR design; G-11	Define a re-issue and revocation policy and test one scan on a phone	Closes a known weak point	Policy decision belongs to the programme owner

Each recommendation is tied to a stakeholder in Chapter 6 §6.3 and to the limitation that motivates it.

4. Future work: split by kind, not by convenience

Category	Items	Why it is future work, not a gap
Existing functionality	Blocking, calibration, score explanation, flag budget	The current scope delivers screening and adjudication; these improve it
Additional capability	Multi-registry operation, shared identity across programmes, confirmation of receipt	Never in the approved scope
ML	Real-data evaluation, retraining pipeline, drift monitoring, cost-sensitive threshold selection	Blocked by the absence of labelled data
Security	Penetration test, secret rotation, key management, rate-limit tuning	Assurance work, outside the project's testing scope
Scale and operations	Load testing, monitoring, backup schedule, high availability	Requires a production budget
Research	Comparing blocking strategies; reviewer efficiency against a manual baseline	Needs a field setting

Not to be listed here: the functional gaps FR-34…FR-37 and the deviations FR-31…FR-33. Those were in scope and are reported as missed or altered in §6.2.9.

5. Boundaries the conclusion must keep

Boundary	What it excludes
Data	Every metric is synthetic; nothing is claimed about Kenyan bursary records
Users	No student, officer, guardian or school participated
Scale	Nothing above 50 candidate pairs was timed
Device	Chrome only; no phone camera; no physical printer
Operation	Demonstration instance; no monitoring, backup schedule or high availability
Security	Control inventory, not an assurance statement

Boundary	What it excludes
Governance	Five owner decisions unanswered; no supervisor finding can be closed without its wording

6. Open items affecting Chapter 6

Item	Effect	Owner
G-21 classifier deviation	Must be stated in §6.2.4, not omitted	Author
G-06 user acceptance	§6.4.9 argues future work from an absence, which must be admitted	Author
Exact Chapter 1 objective wording	§6.2.3–§6.2.6 must answer the originals	Author

7. Related files

`CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORKS.md` (the chapter),
`OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md` (objective → requirement → test → conclusion),
`SBMS_EVIDENCE_PROVENANCE_AND_CLAIM_LIMITS.md` (claim limits).